PHY-112 | Principles of Physics-2

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Spring 2025 | Class #6

Department of Mathematics & Natural Sciences

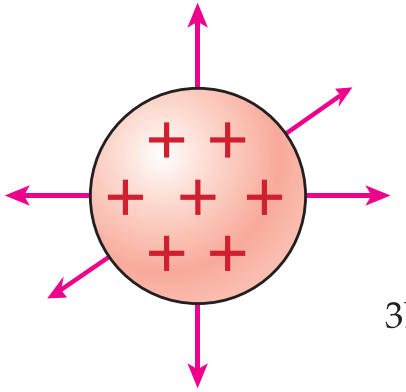
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Enter Volume Charge

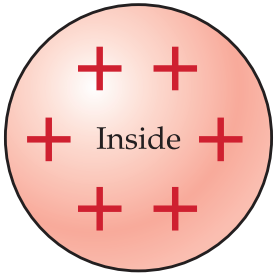
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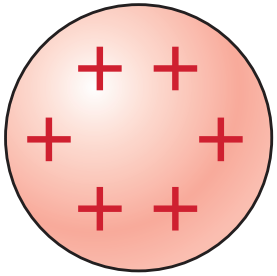
$\vec{E}$ Field for a Spherical Charge

Charge is distributed uniformly over a volume



3D view

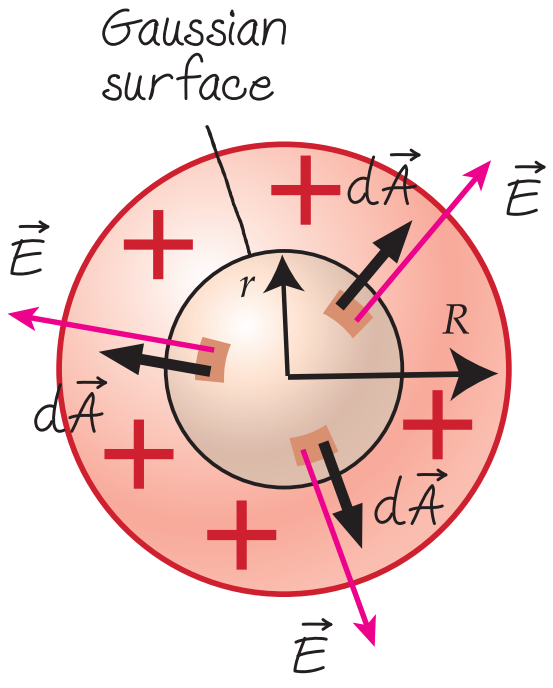




Outside

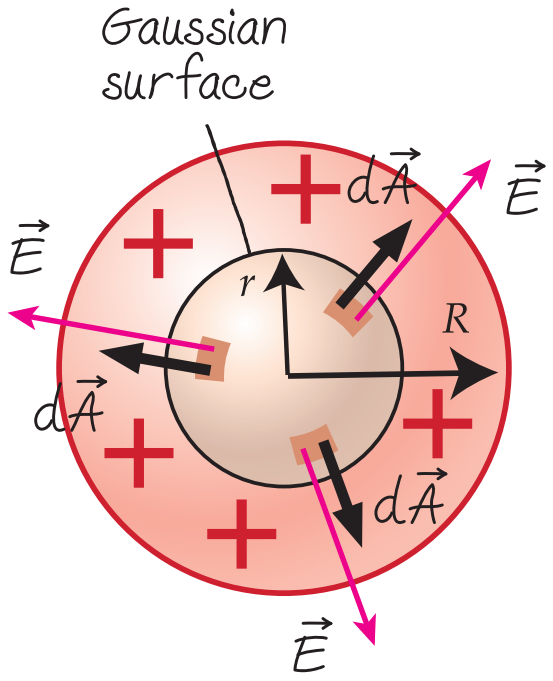
$\vec{E}$ Field for a Spherical Charge: Inside

Charge is distributed uniformly over a volume



$\vec{E}$ Field for a Spherical Charge: Inside

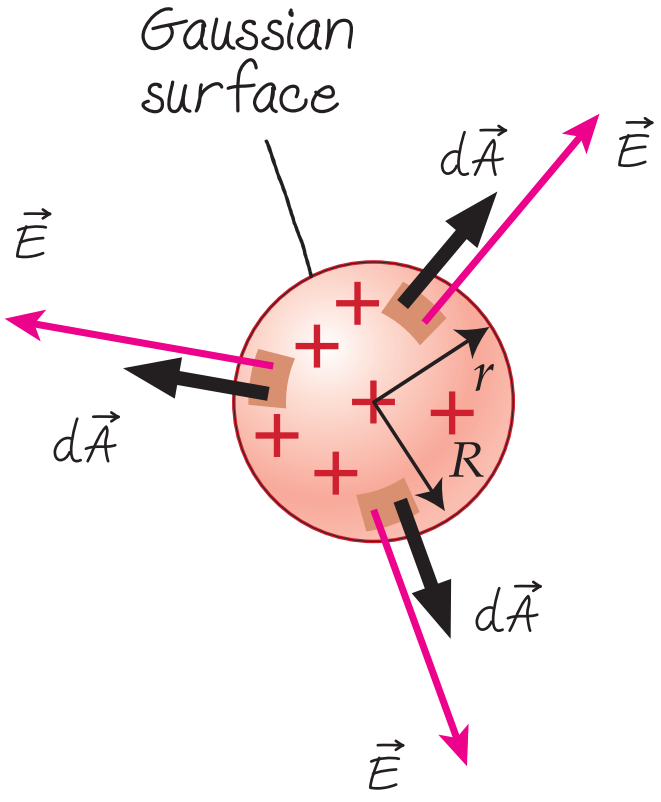
Charge is distributed uniformly over a volume



$$\vec{E}_{\text{inside}} = \left(\frac{qr}{4\pi\epsilon_0 R^3}\right)\hat{r}$$

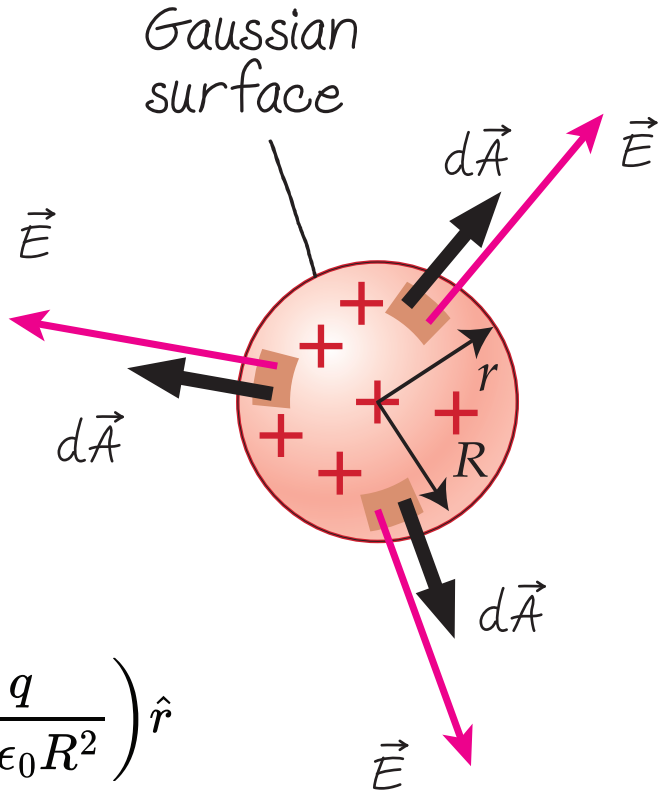
$\vec{E}$ Field for a Spherical Charge: On surface

Charge is distributed uniformly over a volume



$\vec{E}$ Field for a Spherical Charge: On surface

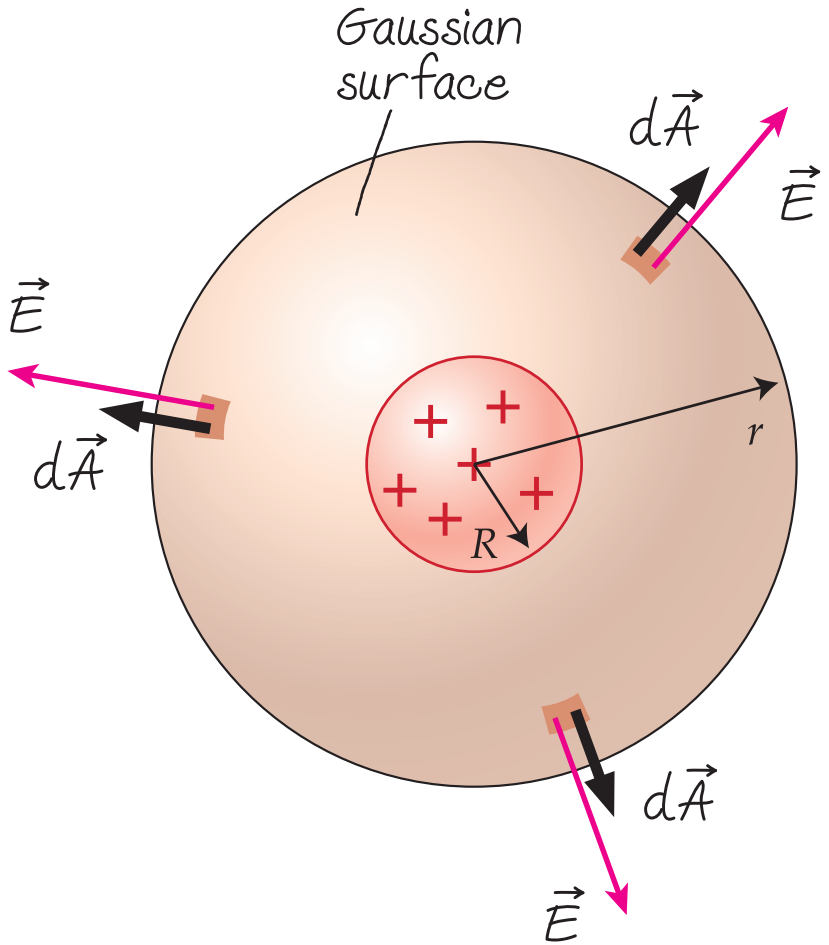
Charge is distributed uniformly over a volume



$$\vec{E}_{\text{on-surface}} = \left(\frac{q}{4\pi\epsilon_0 R^2}\right)\hat{r}$$

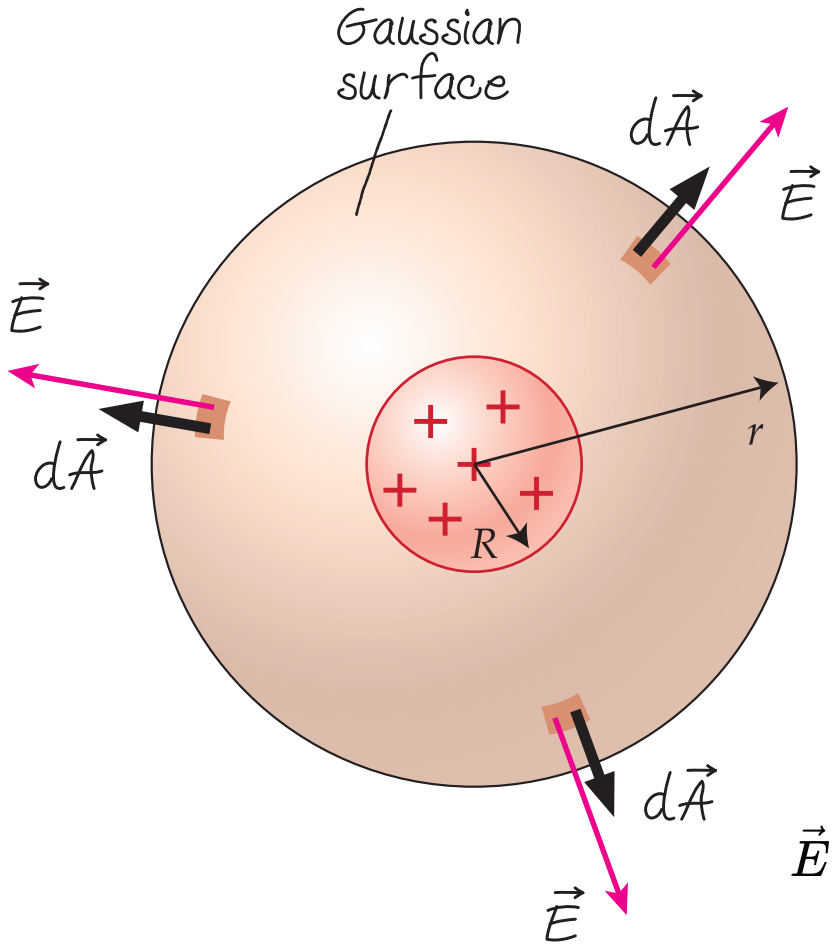
$\vec{E}$ Field for a Spherical Charge: Outside

Charge is distributed uniformly over a volume



$\vec{E}$ Field for a Spherical Charge: Outside

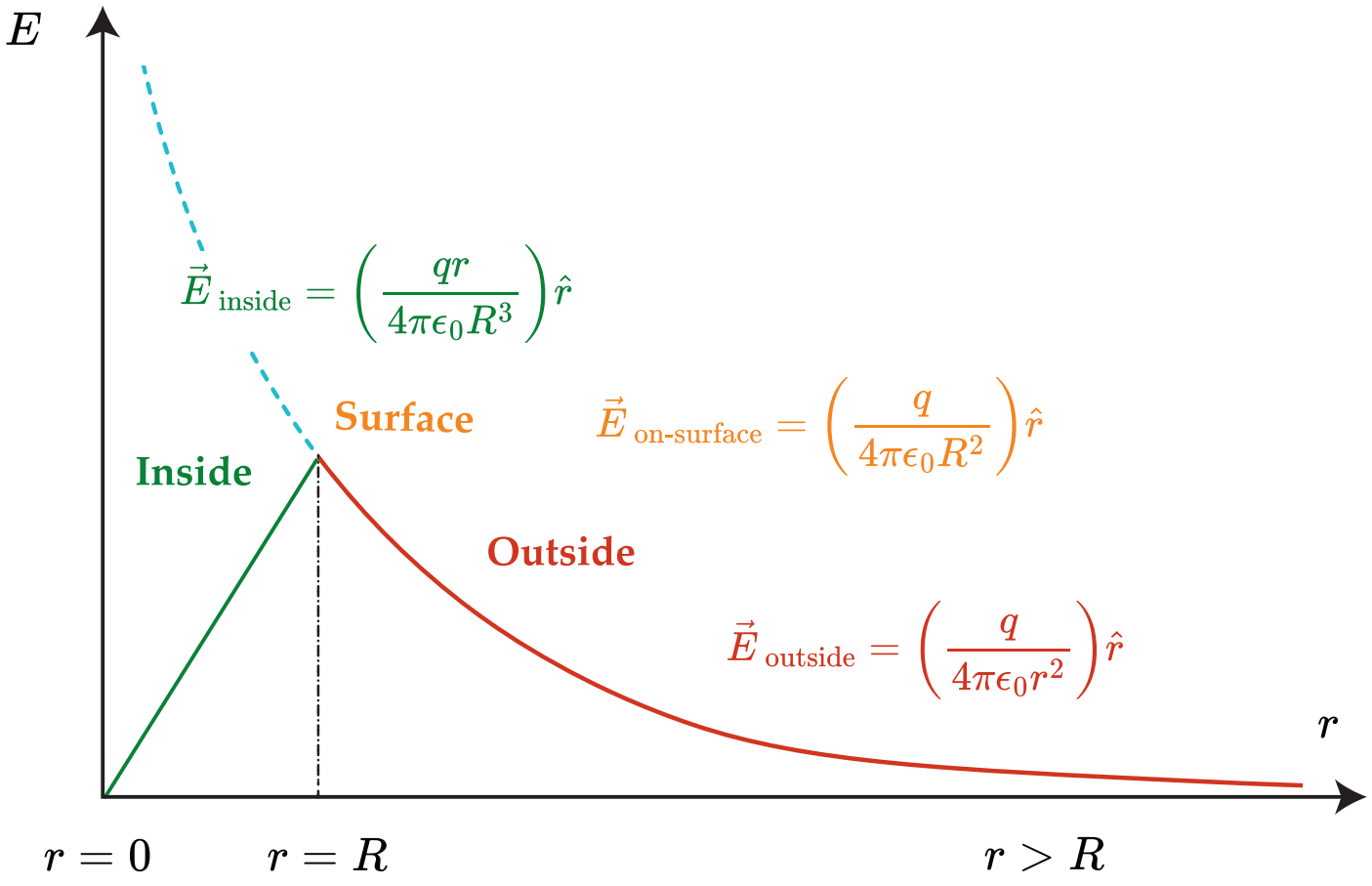
Charge is distributed uniformly over a volume



$$\vec{E}_{\text{outside}} = \left(\frac{q}{4\pi\epsilon_0 r^2}\right)\hat{r}$$

$\vec{E}$ Field for a Spherical Charge

Gauss's Law at work where Coulomb failed!



$\vec{E}$ Field for 3 Model Sources Using Gauss's Law

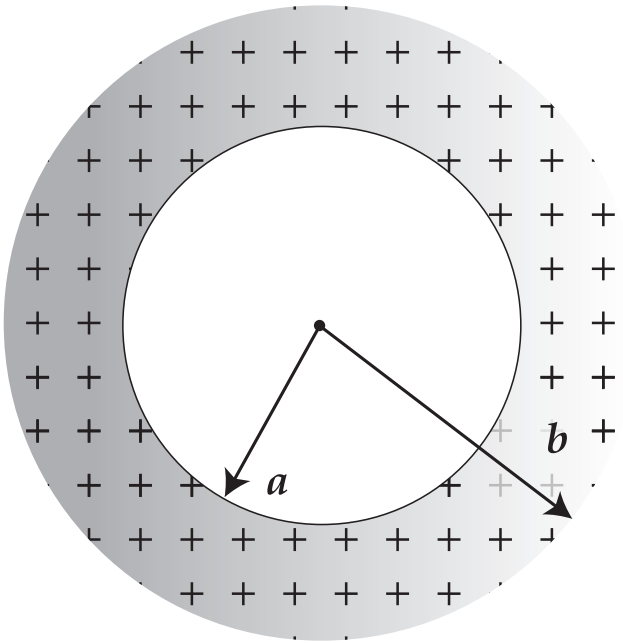
Glance over it at once

- Line Charge $\longrightarrow \vec{E} = \left(\frac{\lambda}{2\pi\epsilon_0 r}\right)\hat{r}$; with $\lambda = \frac{q}{L}$.
- Surface Charge $\longrightarrow \vec{E} = \left(\frac{\sigma}{2\epsilon_0}\right)\hat{n}$; with $\sigma = \frac{q}{A}$.
- Point Charge $\longrightarrow \vec{E} = \left(\frac{q}{4\pi\epsilon_0 r^2}\right)\hat{r}$; with $q = Q_{enc}$.
- Volume Charge $\longrightarrow \vec{E} = \begin{cases} \left(\frac{qr}{4\pi\epsilon_0 R^3}\right)\hat{r} = \left(\frac{r\rho}{3\epsilon_0}\right)\hat{r} & \text{(Inside)} \\ \left(\frac{q}{4\pi\epsilon_0 R^2}\right)\hat{r} & \text{(On Surface)} \\ \left(\frac{q}{4\pi\epsilon_0 r^2}\right)\hat{r} & \text{(Outside)} \end{cases}$

Incepting Ideas (3)

Hint: Think spherical. Adjust Your Gaussian surface accordingly

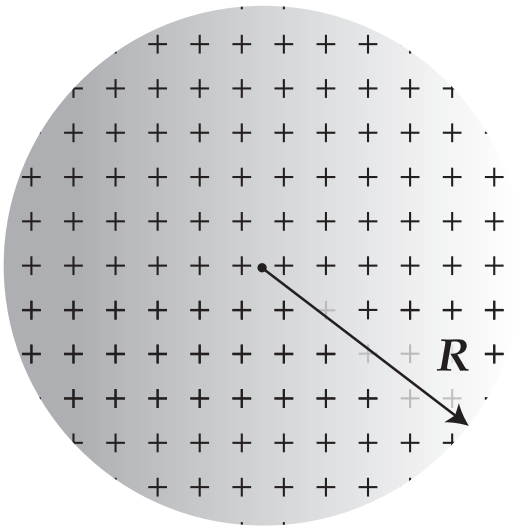
Q: A spherical shell with uniform volume charge density $\rho = 1.84\ \text{nC m}^{-3}$, inner radius $a = 10.0$ cm, and outer radius $b = 2.00a$. (i) How much charge is distributed in the shell? What are the magnitudes and directions of the electric field at radial distances (ii) $r = 0$, (iii) $r = 0.50a$, (iv) $r = a$?



Incepting Ideas (4)

Hint: Think spherical. Adjust Your Gaussian surface accordingly

Q: A spherical charge distribution with uniform volume charge density $\rho = 1.84\ \text{nC m}^{-3}$, with radius $R = 20.0$ cm. (i) How much charge is distributed within this shape? What are the magnitudes and directions of the electric field at radial distances (ii) $r = 0$, (iii) $r = 0.50R$, (iv) $r = R$, (v) $r = 1.5R$, and (vi) $r = 3b$?



Incepting Ideas (5)

Hint: Think spherical. Adjust Your Gaussian surface accordingly

Q: The electric field at a distance of 0.145 m from the surface of a solid insulating sphere with radius 0.355 m is 1750 N C^{-1}. (i) Assuming the sphere's charge is uniformly distributed, what is the charge density inside it? (ii) Calculate the electric field inside the sphere at a distance of 0.200 m from the center.

Similar problem to look at for Additional Practice at home

Exercises 22.43, p-772 | Young-Freedman
Exercise Problem 51, 65, p-683 | Resnick-Halliday

Incepting Ideas (6)

Hint: Do not forget to flux!

Q: A charged particle causes an electric flux of $-750\ \mathrm{N\,m^2\,C^{-1}}$ to pass through a spherical Gaussian surface of 10.0 cm radius centered on the charge. (i) What is the charge of the particle? (ii) If the radius of the Gaussian surface were doubled, how much flux would pass through the surface?

Incepting Ideas (7)

Hint: Hollow shell holds nothing inside

Homework Practice Problem: Try it Yourself

Q: Charge Q is uniformly distributed in a sphere of radius R. (i) What fraction of the charge is contained within the radius $r = 0.50R$? (ii) What is the ratio of the electric field magnitude at $r = 0.50R$ to that on the surface of the sphere? (Exercise Problem 65, p-683 | Resnick-Halliday)

Q: A spherical conducting shell has a charge of -14 μC on its outer surface and a charged particle in its hollow. If the net charge on the shell is 10 μC, what is the charge (i) on the inner surface of the shell and (ii) of the particle? (Exercise Problem 77, p-683 | Resnick-Halliday)

Introducing the 1st Maxwell Equation

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Maxwell's Equation #1 in its full glory

Spoiler Alert: It is Gauss's law for Electric Fields

- The total Φ_E passing through a closed surface is proportional to the total electric charge Q_{enc} enclosed within that surface

$$\oint \vec{E} \cdot d\vec{a} = \frac{Q_{\text{enclosed}}}{\varepsilon_0}. \qquad \text{(Integral Form)}$$

- The Divergence of $\vec{E}$-fields through a closed surface is directly proportional to the charge distribution within that surface

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho_{\text{enclosed}}}{\epsilon_0}. \qquad \text{(Differential Form)}$$

Gauss's Divergence Theorem

From Surface to Volume

- Surface Integral (Vector Field) or Flux (Vector Field) = Volume Integral (Divergence of the same Vector Field)

$$\oint_S \vec{V} \cdot d\vec{a} = \int_V (\vec{\nabla} \cdot \vec{V}) dV$$

$$\oint_S \vec{E} \cdot d\vec{a} = \int_V (\vec{\nabla} \cdot \vec{E}) dV$$

That is it for today!

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